

Genre Classification Using the Spotify Tracks Dataset

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1 Project Idea / Design Questions

This project focuses on multi-class music genre classification using structured audio features from the Spotify Tracks Genre dataset. The goal is to evaluate how effectively traditional machine learning models can predict a song's genre based solely on numerical acoustic characteristics.

Design questions include:

- Which machine learning model performs best for multi-class genre classification?
- How do different audio features contribute to genre prediction?
- How do Logistic Regression, Random Forest, and Support Vector Machines compare in terms of accuracy and generalization?

2 Dataset Details

We used the **Spotify Tracks Genre Dataset** available on Kaggle:

<https://www.kaggle.com/datasets/thedevastator/spotify-tracks-genre-dataset>

The dataset contains approximately 114,000 tracks across 125 distinct genres. Each instance represents a single track and includes numerical audio features extracted from Spotify's API.

Key features include:

- danceability
- energy
- loudness
- speechiness
- acousticness
- instrumentalness
- liveness
- valence
- tempo

The target variable is `track_genre`, making this a supervised multi-class classification problem.

To reduce the complexity of the classification problem, we limited our analysis to the top 20 most frequent genres in the dataset. This helped improve model performance and made evaluation more manageable. The selected top 20 genres are approximately balanced, with each genre containing around 1000 tracks. This helps ensure that model performance is not biased toward any particular class.

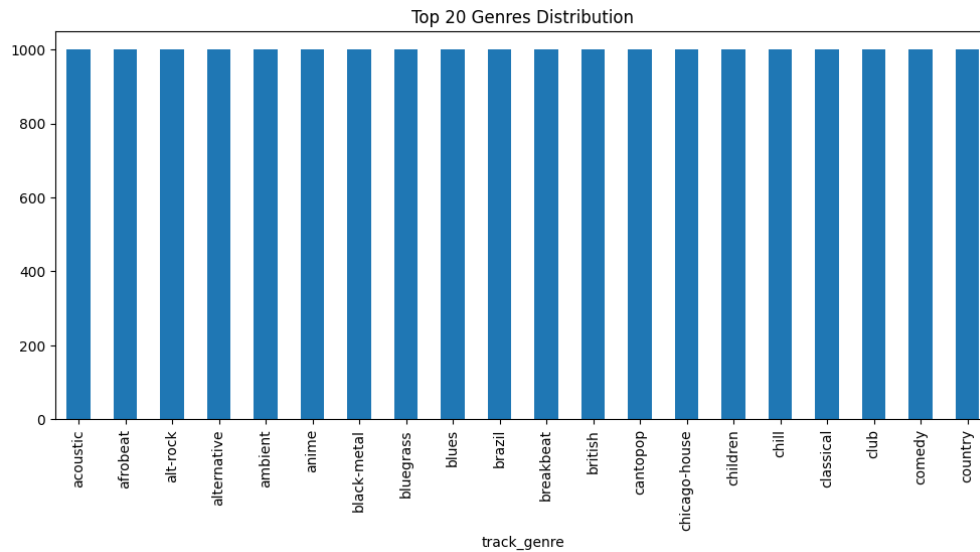


Figure 1: Distribution of the top 20 genres used in the project

3 Proposed Machine Learning Task

The task is supervised multi-class classification.

Input: Numerical audio features

Output: Predicted genre label

We implemented and compared:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)

4 Software

We implemented the project in Python using Google Colab. Core libraries include:

- **pandas**, **numpy** for data loading, cleaning, and numerical operations
- **scikit-learn** for model training and evaluation, preprocessing, and metrics
- **matplotlib** for plots and visualizations
- **imbalanced-learn** for handling class imbalance

We wrote code to:

- load and preprocess the dataset (handle missing values, remove duplicates if needed, encode labels)
- split data into training/validation/test sets
- standardize features
- train and tune Logistic Regression, Random Forest, and SVM models
- evaluate models using multiple metrics and generate plots/tables for the report

5 Model Development and Results

We implemented and compared four models: a baseline Dummy Classifier, Logistic Regression, Random Forest, and Support Vector Machine (SVM).

To evaluate performance, we used accuracy and macro-averaged F1 score, which provides a balanced measure across all classes in this multi-class classification problem.

The results are shown in Table 1.

Model	Accuracy	Macro F1
Baseline	0.0500	0.0048
Logistic Regression	0.3705	0.3475
SVM	0.4123	0.3992
Random Forest	0.4865	0.4803
Tuned Random Forest	0.4865	0.4803

Table 1: Model performance comparison

Among the evaluated models, Random Forest achieved the best performance, with an accuracy of 48.65% and a macro F1 score of 0.48. This indicates that it was the most effective at capturing patterns in the data across multiple genres. The relatively higher performance of Random Forest suggests that nonlinear relationships between audio features play a significant role in genre classification, which linear models like Logistic Regression cannot fully capture.

The Support Vector Machine (SVM) performed moderately well, achieving higher performance than Logistic Regression but lower than Random Forest. Logistic Regression showed the weakest performance among the trained models, likely due to its assumption of linear relationships between features and target classes.

The baseline model performed significantly worse than all other models, with only 5% accuracy, demonstrating that the machine learning models were able to learn meaningful patterns beyond simple guessing.

Hyperparameter tuning did not improve the Random Forest model’s performance, as the tuned model achieved identical results to the default configuration. This suggests that the default parameters were already well-suited for this dataset or that the tuning search space was limited.

6 Confusion Matrix Analysis

To further evaluate the performance of the best model, we examined the confusion matrix for the tuned Random Forest classifier.

The confusion matrix provides insight into how well the model distinguishes between different genres by showing the number of correct and incorrect predictions for each class.

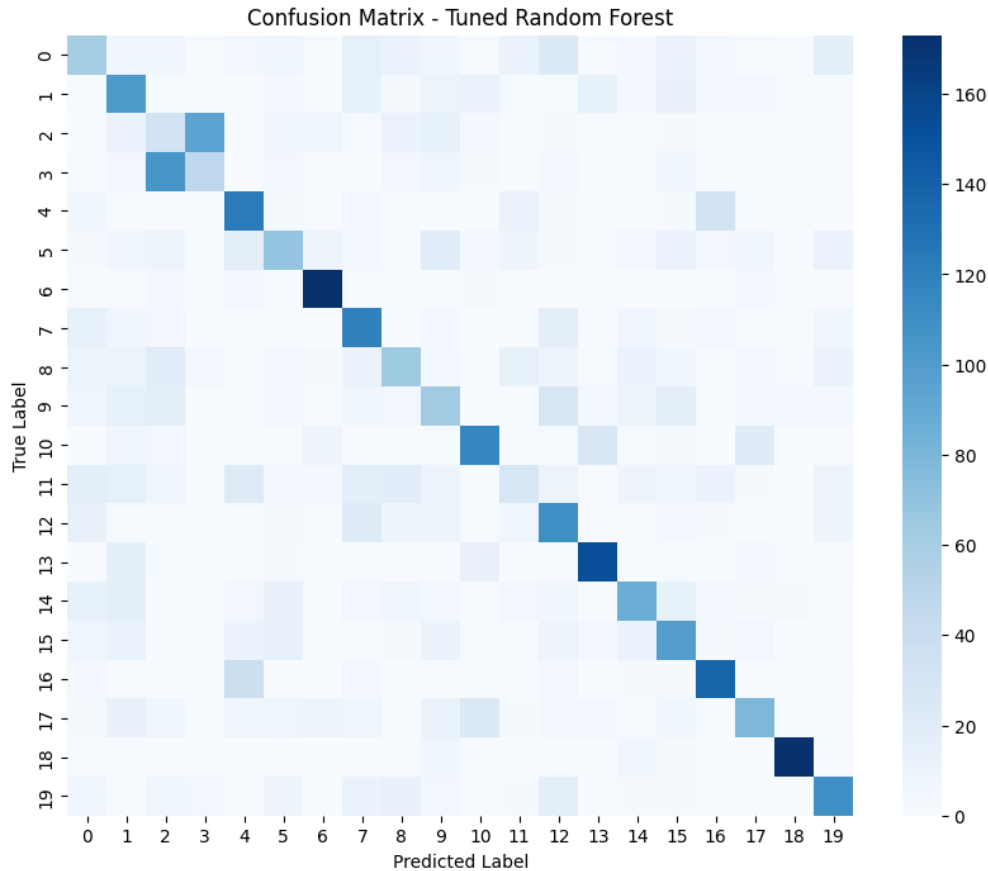


Figure 2: Confusion matrix for the tuned Random Forest model

The confusion matrix shows that the model performs well on some genres but struggles to distinguish between others. This is expected because many music genres share similar audio characteristics, leading to overlap between classes.

Some genres are predicted more accurately, while others are frequently misclassified. This indicates that certain genres have more distinct audio patterns, while others are more similar and difficult to separate.

Overall, the confusion matrix highlights the challenges of multi-class genre classification and supports the observed performance metrics.

7 Feature Importance

To better understand how the model makes predictions, we analyzed feature importance using the tuned Random Forest model.

Feature importance measures how much each input feature contributes to the model's predictions. The results are shown in Figure 2.

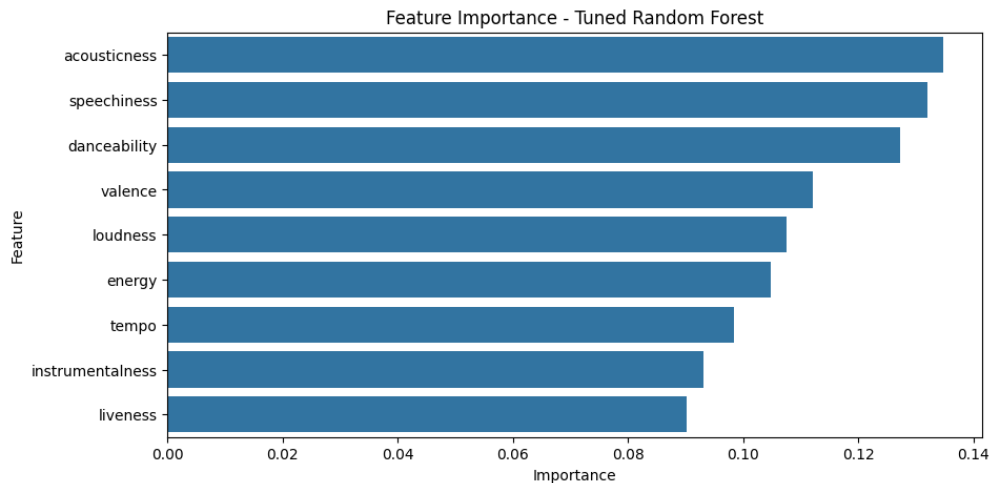


Figure 3: Feature importance from the tuned Random Forest model

The results show that acousticness, speechiness, and danceability were the most important features for predicting music genres.

Acousticness measures whether a track is acoustic or electronic, which plays a major role in distinguishing between genres. Speechiness captures the presence of spoken words, which helps identify genres such as rap or spoken content. Danceability reflects rhythm and beat, which are key characteristics across many music genres.

Other features such as valence, loudness, energy, and tempo also contributed to the model, but to a slightly lesser extent. The relatively balanced importance values suggest that multiple audio features work together to determine a song's genre.

8 Limitations

One limitation of this project is that even after reducing the dataset to the top 20 genres, there is still significant overlap between genres. Many genres share similar audio characteristics, which makes classification difficult and reduces overall accuracy.

Additionally, only numerical audio features were used in this project. While these features capture important aspects of music, they may not fully represent the complexity of each genre. Including additional information such as lyrics, artist metadata, or raw audio features could improve performance.

Another limitation is that the models were trained using a limited set of hyperparameters. Expanding the search space or using more advanced tuning methods could potentially improve the results.

9 Conclusion

In this project, we compared multiple machine learning models for multi-class music genre classification using audio features from the Spotify Tracks dataset.

Among the models tested, Random Forest achieved the best performance, demonstrating its ability to capture complex relationships between features. Other models such as SVM and Logistic Regression performed reasonably well but were less effective.

The results show that audio features such as acousticness, speechiness, and danceability play an important role in distinguishing between genres. However, the overall performance indicates that genre classification remains a challenging problem due to similarities between classes.

In future work, more advanced models or additional features could be explored to improve classification performance.